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Floer Homology (WS 2021/22)
Exercises

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Sheet 1

Autonomous vector fields and flows

Definiton. A smooth (autonomous/time-independent) vector field X on smooth n-manifold M is a smooth section in $\Gamma(TM)$.

Local expression: Around each point $p \in M$, there is a local chart $\overset{\varphi: U \to \mathbb{R}^n}{p \mapsto x=(x_1,\dots,x_n)}$ in which there is unique expression $X = \sum_{i=1}^n v_i \frac{\partial}{\partial x_i}$ where $v_i \in C^\infty(U)$ for $i = 1, \dots, n$.

Definition. A flow on M is a smooth map $\phi : M \times \mathbb{R} \to M$ ($\phi_t := \phi(\cdot, t)$) satisfying:

i) $\phi_0 = \mathrm{id}_M$.

ii) $\phi_t \circ \phi_s = \phi_{t+s}$ $\forall\, t, s \in \mathbb{R}$.

Theorem (Existence & Uniquness). Given $X \in \Gamma(TM)$, for every $p \in M$, $\exists\, \varepsilon > 0$ and unqqiue smooth curve $\phi_t(p) : (-\varepsilon, \varepsilon) \to M$ solving

$$(\star) \qquad \begin{cases} \frac{d}{dt}\phi_t(p) = X(\phi_t(p)) \\ \phi_0(p) = p \end{cases}$$

The solution depends smoothly on p. The smooth map $\phi : \mathcal{D} \to M$ is a local flow ($\mathcal{D} \subseteq M \times \mathbb{R}$ open subset is its maximal defining domain).

Terminology: A vector field X is complete if it generates a global flow ϕ (i.e. $\mathcal{D} = M \times \mathbb{R}$).

Lemma. If the maximal local flow ϕ of X contains an element of the form $\phi : (-\varepsilon, \varepsilon) \times M \to M$ for some $\varepsilon > 0$, then X is complete.

Theorem. If X is compact supported (e.g. M is compact), then it is complete.

Exercise.

(a) Show the vector fields $X = x^2 \frac{\partial}{\partial x}$ and $X = e^x \frac{\partial}{\partial x}$ are not complete on $\mathbb{R}$.

(b) Every left-invariant vector field on a Lie group is compelte.

Sheet 2

Time-dependent Hamiltonians and Hamiltonian isotopies

On symplectic manifold (M, ω):

Time-dependent Hamiltonian: $H \in C^\infty(M \times \mathbb{R})$, $H_t := H(\cdot, t)$.

Hamiltonian vector field: $X_{H_t} \in \Gamma(TM)$, $i_{X_{H_t}}\omega = -dH_t \ \forall\, t \in \mathbb{R}$ (Noter: Hamiltonian vector field is uniquely dtermined by Hamiltonian due to the bundle isomorphism induced by symplectic structrue $\omega : TM \xrightarrow{\cong} T^*M$).

Hamiltonian isotopy based at identity: $\phi_t^H \in \mathrm{Diff}(M)$, $\phi_0^H = \mathrm{id}_M$, $\frac{d}{dt}\phi_t^H = X_{H_t} \circ \phi_t^H$.

Notation: $\mathrm{Symp}(M, \omega)$ the group of all symplectomorphisms, $\mathrm{Ham}(M, \omega)$ the set of all Hamiltonian diffeomorphism.

It is easy to see that $\mathrm{Ham}(M, \omega) \subseteq \mathrm{Symp}_0(M, \omega)$.

Goal: Show that $\mathrm{Ham}(M, \omega)$ actually form a group (equipped with smooth structure, hence a Lie group).

Lemma. Given $\phi \in \mathrm{Symp}(M, \omega)$ and $H \in C^\infty(M)$, $X_{H \circ \phi^{-1}} = \phi_* X_H$.

Proposition. Let ϕ_t, ψ_t be Hamiltonian isotopies generated by Hamiltonian H_t, F_t. Then the Hamiltonian isotopy $\theta_t := \phi_t \circ \psi_t$ is generated by Hamiltonian $K_t := H_t + F_t \circ \phi_t^{-1}$.

Corollary. If $\phi \in \mathrm{Symp}(M, \omega)$ is Hamiltonian, so is ϕ^{-1}.

Consequence: $\mathrm{Ham}(M, \omega)$ is indeed a group.

It is now not hard to see that $\mathrm{Ham}(M, \omega) \triangleleft \mathrm{Symp}(M, \omega)$.

Terminology: H_t is said to be normalized if

- When M is closed: $\int_M H_t \omega^n = 0 \ \forall\, t$.
- When M is open: $\bigcup_t \mathrm{supp}\, H_t \subset K$ where $K \subset M$ is some compact subset.

Theorem (Banyaga). Every Hamiltonian isotopy based at id_M is generated by some normalized Hamiltonian.

Exercise. Suppose F, G are Hamiltonians on (M, ω), ϕ_t, ψ_t are the corresponding Hamiltonian isotopies.

a) The Poisson bracket between Hamiltonians is defined as $\{F, G\} := \frac{d}{dt}\big|_{t=0} \mathrm{Ad}_{\phi_t} G := \frac{d}{dt}\big|_{t=0} \phi_t \psi_t \phi_t^{-1}$ (Recall: $\mathrm{Ad}_\phi G := \frac{d}{dt}\big|_{t=0} \phi \psi_t \phi^{-1}$). Show $\{F, G\} = \omega(X_G, X_F)$.

b) Show $[X_F, X_G] = -X_{\{F,G\}}$.

c) Show that: if $\phi_t \circ \psi_t = \psi_t \circ \phi_t \; \forall t$, then $\{F, G\} = 0$.

d) For $H, g \in C^\infty(M)$, show that g is invariant under the Hamiltonian flow of H iff. $\{H, g\} = 0$ (Such g is called an "integral of motion" w.r.t. the autonomous Hamiltonian H).

Consequence: The Lie algebra of $\mathrm{Ham}(M, \omega)$ (i.e. Hamiltonian vector fields w.r.t. $[\cdot, \cdot]$) is isomorphic to the Lie algebra consits of normalized Hamiltonian with $\{\cdot, \cdot\}$.

Exercise. Suppose $H^1_c(M; \mathbb{R}) = 0$. Then $\mathrm{Ham}(M, \omega) = \mathrm{Symp}_0(M, \omega)$.

Hint: Apply Cartan's magic formula.

Sheet 3

Almost complex structures

Definition. A complex structrue on a real vector space is an edomorphism $J \in \mathrm{End}(V)$ s.t. $J^2 = -\mathrm{id}$. On a symplectic vector space (V, ω) (recall: ω is a non-degenerate, skew-symmetric bilinear form), a complex structrue is said to be ω-compatible if $\omega(\cdot, J\cdot)$ is a positive definite symmetric bilinear form (i.e. inner product).

Remark.

1) (V, J) can be tought of as a complex vector space via complexification $(a+\mathbf{i}b)v = av + bJv \ \forall\, v \in V$.

2) Being symmetric is equivalent to $\omega(J\cdot, J\cdot) = \omega(\cdot, \cdot)$.

Lemma. On (V, ω), ω-compatible J always exists.

Proposition. Given a inner product on (V, ω), one can construct an ω-compatible complex structrue J.

Remark. $\{$Euclidean structures on $V\} \overset{1:1}{\longleftrightarrow} \{\omega$-compatible complex structures on $(V, \omega)\}$

Definition. An almost complex structure on a manifold M is $J \in \mathrm{End}(TM)$ s.t. $J^2 = -\mathrm{id}_{TM}$ (i.e. J_x is a complex structure on $T_xM \ \forall\, x \in M$). On symplectic manifold (M, ω), an almost complex structure is said to be ω-compatible if $g_J(\cdot, \cdot) = \omega(\cdot, J\cdot)$ is a Riemannian metric.

Fact: In the compatible triple (ω, g, J), any two determie the third:

- $g(\cdot, \cdot) = \omega(\cdot, J\cdot)$
- $\omega(\cdot, \cdot) = g(J\cdot, \cdot)$
- $J(\cdot) = g^{-1}(\omega(\cdot))$ (here g, ω are thought of as the induced isomorphisms $TM \overset{\cong}{\to} T^*M$)

Lemma. On (M, ω, g), there exsits a canonical almost complex structure J compatible with ω.

Corollary. Any symplectic manifold has compatible almost complex structure.

Notation: Denote the space of compatible almost complex structrue $\mathcal{J}(M, \omega)$.

Proposition. $\mathcal{J}(M, \omega)$ is path-connected, and further contractible.

Proposition. On an almost complex manifold (M, J), the space of J-compatible symplectic structures ω is convex (hence contractible).

Note: Such space of symplectic structure might be empty at the very begining, since there are almost complex manifolds admiting no symplectic structures, e.g. $\mathbb{S}^6$.

Lemma. On a general manifold, $\exists$ an almost complex structure $\Leftrightarrow$ $\exists$ a non-degenertate 2-form (aka. almost symplectic structure).

Note: This is by no mean a 1-1 corrrespondence, but simply an equivalence of exsitence.

Theorem (Borel-Serre). $\mathbb{S}^n$ carries an almost complex structure iff. $n = 2$ or 6.

Exercise. Show that $\mathbb{S}^{2k}$ $(k \geq 1)$ carries a symplectic structure iff. $k = 1$.

Exercise. Show that $\mathbb{S}^{4k}$ $(k \geq 1)$ does not admit any almost complex structure.

Hint: First show that if there is an almost complex structure on $\mathbb{S}^{4k}$, then $(-1)^k p_k(T\mathbb{S}^{4k}) = 2e(T\mathbb{S}^{4k})$.

Sheet 4

Complex structures and Kähler structures

Definition. An almost complex structure J on M is said to be integrable if it comes from a complex manifold structure, i.e. (M, J) admits atlas s.t. it is locally holomorphic to $(\mathbb{C}^n, \mathbf{i})$. An almost complex structure is a complex structure iff. it is integrable.

Definition (Nijenhuis tensor for J). $\forall\, X, Y \in \Gamma(TM)$:

$$\mathcal{N}_J(X,Y) := [JX, JY] - J[X, JY] - J[JX, Y] - [X, Y].$$

Theorem (Newlander-Nirenberg). J is integrable $\Leftrightarrow \mathcal{N}_J \equiv 0$.

Remark. "$\Leftarrow$" is the difficult part!

Exercise.

a) Show that $\mathcal{N}_J \equiv 0 \Leftrightarrow [T^{1,0}M, T^{1,0}M] \subset T^{1,0}M$.

Hint: Show $\mathcal{N}_J(X,Y) = -8\,\mathrm{Re}([X^{1,0}, Y^{1,0}]^{0,1})$.

b) Show that if J is integrable, then $\mathcal{N}_J \equiv 0$ ("$\Rightarrow$" part in Newlander-Nirenberg thm.).

c) Show that every almost complex structure on any surface is integrable (This implies all Riemann surfaces are Kähler).

Definition. A symplectic manifold (M, ω) is Kähler if it admits a integrable ω-compatible almost complex structrue J. In other words, (M, J) is a complex manifold carrying a closed positive non-degenerate real $(1,1)$-form ω.

Note: If J is merely an almost complex structure, then (M, ω, J) is called an almost Kähler manifold.

Fact: Kähler manifold (M, ω, J) has $b_{2k+1}(M) \in 2\mathbb{Z}\ \forall\, k \in \mathbb{N}_0$.

Exercise. William Thurston discovered in 1976 an example that is symplectic but not Kähler: Consider $\mathbb{Z}^4$ (as a set) equipped with the non-commutative group structure: $a * b = a + L_a b\ \forall\, a, b \in \mathbb{Z}^4$ where

$$L_a = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & a_2 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

(i.e. one can show the group law is satisfied: $L_a L_b = L_{a+b} = L_{a*b}$, $a^{-1} = -L_{-a}a$.) The group $G := (\mathbb{Z}^4, *)$ acts on $\mathbb{R}^4$ by $\overset{G\times\mathbb{R}^4\to\mathbb{R}^4}{(a,x)\mapsto a+L_a x}$, this smooth action is free and proper s.t. $N := \mathbb{R}^4/G$ is a smooth manifold. Find a symplectic structrue on N and show that N does not admits any Kähler structrue.

Hint: Apply Hurewicz's thm. and show that $G^{\mathrm{Ab}} := G/[G,G] \cong (\mathbb{Z}^3, +)$.

Sheet 5

Loop and path spaces

Definition. The free loop space of a topological space X is the space

$$\mathcal{L}X := \mathrm{Map}(\mathbb{S}^1, X) := \{f : \mathbb{S}^1 \to X \text{ continuous}\}$$

equipped with compact-open topology. In category theory, it is denoted as $\mathrm{Hom}_{\mathsf{Top}}(\mathbb{S}^1, X)$. In smooth category, i.e. X is a smooth manifold, it is usually denoted as $C^0(\mathbb{S}^1, X)$.

Definition. The based loop space of X is the space

$$\Omega X := \mathrm{Map}((\mathbb{S}^1, s_0), (X, x_0)) := \{f : (\mathbb{S}^1, s_0) \to (X, x_0) \text{ continuous} \,|\, f(s_0) = x_0\}$$

equipped with conpact-open topology. Specifying the base point, it is then denoted as $\Omega_{x_0} X$.

Definition. The path space of X is the space

$$\mathcal{P}X := \mathrm{Map}((I, \{0\}), (X, x_0)) := \mathrm{Map}(I, X) := \{f : [0,1] \to X \text{ continuous} \,|\, f(0) = x_0\}$$

equipped with compact-open topology. When specifying the based point, it is then denoted as $\mathcal{P}_{x_0} X$. More generally, w.r.t. subsets $U, V \subset X$ (U, V need not to be distinct), the relative path space is defined as

$$\mathcal{P}_{U,V} X := \{f : [0,1] \to X \text{ continuous} \,|\, f(0) \in U,\ f(1) \in V\}.$$

When $V = U$, then denote the relative path space simply as $\mathcal{P}_U X$.

There are following homotopy pullbacks:

$$\begin{array}{ccc} \mathcal{L}X & \longrightarrow & \mathrm{Map}(I, X) \\ {\scriptstyle \mathrm{ev}}\downarrow & & \downarrow{\scriptstyle \mathrm{ev}_0 \times \mathrm{ev}_1} \\ X & \xrightarrow{\triangle} & X \times X \end{array} \qquad \begin{array}{ccc} \mathcal{P}_{U,V} X & \longrightarrow & \mathrm{Map}(I, X) \\ {\scriptstyle \mathrm{ev}_0 \times \mathrm{ev}_1}\downarrow & & \downarrow{\scriptstyle \mathrm{ev}_0 \times \mathrm{ev}_1} \\ U \times V & \xrightarrow{i_U \times i_V} & X \times X \end{array}$$

Fact: The following are fibrations (even further fiber bundles):

$$\mathcal{P}X \hookrightarrow \mathrm{Map}(I, X) \overset{\mathrm{ev}_1}{\twoheadrightarrow} X \qquad \Omega X \hookrightarrow \mathcal{P}X \overset{\mathrm{ev}_1}{\twoheadrightarrow} X \qquad \Omega X \hookrightarrow \mathcal{L}X \overset{\mathrm{ev}}{\twoheadrightarrow} X$$

When X is path-connected, all fibers $\Omega_* X$ are homotopy equivalent due to the property of fibrations.

Exercise. Show that $\mathrm{Map}(I, X) \simeq X$ and ΩX is contractible.

Exercise. Suppose X is path-connected, subsets $U, V \subset X$ are contractible in X. Show that $\mathcal{P}_{U,V}X \simeq \Omega X \times U \times V$.

Exercise. Show that $\pi_{n+1}(X) \cong \pi_n(\Omega X)\ \forall\, n \in \mathbb{N}$ for path-connected space X.

For $\alpha \in \pi_1(X)$, the subspace $\mathcal{L}_\alpha X \subseteq \mathcal{L}X$ stands for the path-connected component containing loops of homotopy class α, similarly we have $\Omega_\alpha X \subseteq \Omega X$.

Fact: All the path-connected component of ΩX are homotopic equivalent.

This is due to the fact that based loop space is a homotopy associative H-space (i.e. a H-group), where the product is given by the concatenation of loops and the homotopy inverse is the loop reversal. One can show that the path-connected components of H-group structure are all homotopic equivalent.

In particular, fibration $\Omega_0 X \overset{i}{\hookrightarrow} \mathcal{L}_0 X \overset{\text{ev}}{\twoheadrightarrow} X$ admits a section $c : X \to \mathcal{L}_0 X$ given by inclusion of constant loops. The monodromy action $\pi_1(X) \to \mathrm{Aut}(\Omega X)$ given by conjugcation $\underset{(\alpha,\gamma)\mapsto \alpha^{-1}*\gamma*\alpha}{\Omega X \times \Omega X \to \Omega X}$, and this induced further an action on the fundamental group of fibers $\underset{(\alpha,\gamma)\mapsto i_*^{-1}(c_*(\alpha) i_*(\gamma) c_*(\alpha)^{-1})}{\pi_1(X)\times\pi_1(\Omega_0 X)\to\pi_1(\Omega_0 X)}$. Under the isomorphism $\pi_1(\Omega_0 X) \cong \pi_1(\Omega X) \cong \pi_2(X)$, this turns out to be the canonical action of $\pi_1(X)$ on $\pi_2(X)$.

Exercise. Let X be path-connected space, proves the following:

a) $\pi_n(\mathcal{L}_0 X) \cong \pi_{n+1}(X) \oplus \pi_n(X)\ \forall\, n \geq 2$.

b) $\pi_1(\mathcal{L}_0 X) \cong \pi_2(X) \rtimes \pi_1(X)$.

Hint: Recall a semi-direct product of groups is equivalent to a split group extension, i.e. a split short exact sequence of groups.

c) There is a set bijection $\pi_0(\mathcal{L}X) \overset{1:1}{\longleftrightarrow} \pi_1(X, x_0)/_{\text{conjugations}}$.

In the category of smooth manifolds, for instance on a finite dimensional smooth manifold M, it is more practical to consider the spaces $C^\infty(\mathbb{S}^1, M), W^{1,2}(\mathbb{S}^1, M)$, etc. Same for path spaces. Thinking of these spaces as infinite dimensional manifolds equipped with norms, we have the well-known results:

Theorem. For finite diemsnional compact connected Riemannian manifold, the continuous inclusion $C^k(\mathbb{S}^1, M) \subseteq C^0(\mathbb{S}^1, M)$ and $C^k(\mathbb{S}^1, M) \subseteq W^{1,2}(\mathbb{S}^1, M)$ are homotopy equivalence when $k \geq 1$. The dense continuous inclusion $W^{1,2}(\mathbb{S}^1, M) \subseteq C^0(\mathbb{S}^1, M)$ is therefore an homotopy equivalence as well.

A straightforward proof can be found in Chapter 1 of *Introduciton to Global Analysis*, Theorem 1.5.1, by J. D. Moore.

Generally, one can show by using smooth approximation method (c.f. M. W. Hirsch's *Differential Topology*, Chapter 2) that for finite dimensional smooth manifold M (need not to be compact) the Hilbert manifolds $W^{k,2}(\mathbb{S}^1, M)$ are weakly

homotopy equivalent to Fréchet manifold $C^\infty(\mathbb{S}^1, M) = \bigcap_{k\geq 1} W^{k,2}(\mathbb{S}^1, M)$ and further to $C^0(\mathbb{S}^1, M)$ via continuous inclusions:

$$C^\infty(\mathbb{S}^1, M) \subset \cdots \subset C^k(\mathbb{S}^1, M) \subset W^{k,2}(\mathbb{S}^1, M) \subset C^{k-1}(\mathbb{S}^1, M) \subset \cdots \subset C^0(\mathbb{S}^1, M).$$

Then by Theorem 15 in *Homotopy theory of infinite dimensional manifolds* by R. S. Palais saying that weakly homotopy equivalence between metrizable manifolds (could be infinite dimensional) is acutally homotopy equivalence, these spaces are homtpy equivalent. Alternatively one can easily see that sequence of continuous inlusions of $W^{k,2}$-class free path spaces are homotopy equivalences:

$$W^{\infty,2}(I, M) \subset \cdots \subset W^{k,2}(I, M) \subset \cdots \subset L^2(I, M).$$

Then by the property that pull-back of weakly homotopy equivalent fibrations are weakly homotopy equivalent, we conclude that the fibrations $W^{k,2}(\mathbb{S}^1, M) \xrightarrow{\text{ev}} M$, regarded as the pull-back of (weakly) homotopy equivalent fibrations

$$W^{k,2}(I, M) \xrightarrow{\text{ev}_0 \times \text{ev}_1} M \times M$$

along $M \xrightarrow{\triangle} M \times M$ (i.e. $W^{k,2}(I, M) \times_{M\times M} M \cong W^{k,2}(\mathbb{S}^1, M)$), are weakly homotopy equivalent. More detail description lies in *Basics on free loop space*, Section 2, by D. Chataur and A. Oancea.

For an up-to-date detail account on applying smooth approximation to show the weakly homotopy equivalence given by sequence of continuous inclusion

$$C^\infty(M, N) \subset \cdots C^k(M, N) \subset C^{k-1}(M, N) \subset \cdots \subset C^0(M, N)$$

w.r.t. weak Whitney topology, where M, N are arbitrary finite dimensional smooth manifolds, one can refer to *Smooth appriximations and their applications to homotopy types* (2020) by O. Khokhliuk and S. Maksymenko.

With above knowledge in mind, we justify why people often define free loop space of smooth manifold M as the space of smooth loops: $\mathcal{L}M := C^\infty(\mathbb{S}^1, M)$. Similar reasons hold for based loop spaces and path spaces.